

Program: Diploma in Civil Engineering	
Course Code: 5358	Course Title: Structural Engineering Drafting Lab
Semester: 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide clear idea about detailing of reinforced concrete and steel structural members corresponding to IS codes.
- To prepare the bar bending schedule of RCC structural members.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of different building materials & methods of construction	3351	Building Materials & Construction	3
Basic concepts of theory of Structures	3014	Theory of structures	3
Basic concepts of building drawing	3358	Civil Engineering Drafting Lab	3
Knowledge of structural design of steel and RCC		Design of Steel and RCC Structures	5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop sketches to detail the RCC beams and slabs.	11	Applying
CO2	Develop the detailed drawings of RCC columns, footings and stairs.	13	Applying
CO3	Develop sketches for the detailing of tension and compression steel members.	11	Applying
CO4	Develop sketches for the detailing of tension and compression steel members.	6	Applying

	Lab Tests	4	
--	-----------	---	--

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3				
CO3	3		3				

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop sketches to detail the RCC beams and slabs.		
M1.01	Show the cross section, strain diagram & stress diagram for singly reinforced section.	1	Understanding
M1.02	Show the cross section, strain diagram & stress diagram for doubly reinforced section.	1	Understanding
M1.03	Determine the sectional properties and sketch the detailing of a singly reinforced beam and prepare bar bending schedule.	3	Applying
M1.04	Determine the sectional properties and sketch the detailing of a one-way slab and prepare bar bending schedule.	3	Applying
M1.05	Develop the reinforcement details of a two-way slab, corners held down and prepare bar bending schedule.	3	Applying
CO2	Develop the detailed drawings of RCC columns, footings and stairs.		
M2.01	Summarize the provisions of IS:456 required for the design of a column.	1	Understanding
M2.02	Model the reinforcement detailing of the circular and square column and prepare bar bending schedule.	3	Applying

M2.03	Determine the sectional properties and sketch the detailing of an isolated column footing and prepare bar bending schedule.	3	Applying
M2.04	Develop the plan and longitudinal section of doglegged staircase and open well staircase showing all the details and dimensions	3	Applying
M2.05	Sketch the detailing of a cantilever retaining wall and prepare its bar bending schedule.	3	Applying
	Lab Test-I	2	
CO3	Develop sketches for the detailing of tension and compression steel members		
M3.01	Summarize the clauses from IS:800 required for the design of a compression member and submit it in report form.	1	Understanding
M3.02	Prepare the sketch of single, double lacing and battening of given built up columns.	3	Applying
M3.03	Model the elevation and cross-section of a plate girder.	3	Applying
M3.04	Develop the sketch for the beam-column connection.	2	Applying
M3.05	Model the elevation and plan of a Gusseted base of a column.	2	Applying
CO4	Develop sketches for the detailing of tension and compression steel members		
M4.01	Open ended experiment- Prepare a detailed report of site visit for reinforcement detailing of structural elements like beams, columns, staircase & footing and prepare bar bending schedule.	6	Applying
	Lab Test-II	2	

Text/Reference:

T/R	BookTitle/Author
T1	Shah, V.L., and Gore, V., Limit State Design of Steel Structures, Structures Publications, Pune.
R2	Dayarathnam, P., Design of Steel Structures, S. Chand and Company, Delhi.
R3	Subramanian N., Design of Steel Structures, Oxford University Press.
R4	Sairam, K.S., Design of Steel Structures, Pearson Publication, Chennai, Delhi.
R5	Shah, V.L., and Karve, S.R., Limit State Theory and Design of Reinforced Concrete Structures, Structures Publications, Pune, 2014.
R6	Sinha N.C., and Roy S.K., Fundamentals of Reinforced Concrete, S. Chand & Co., New Delhi.
R7	Krishna Raju, and N. Pranesh, R.N., Reinforced Concrete Design Principles and Practice, New Age International, Mumbai.

Online Resources:

Sl.No	WebsiteLink
1	https://nptelvideos.com/course.php?id=294
2	https://www.coursera.org/learn/design-basics-of-rcc-buildings
3	https://onlinecourses.swayam2.ac.in/ntr25_ed56/preview
4	https://www.coursera.org/learn/design-basics-of-steel-buildings